

Final Report – Group 7 – ME 5751

Date: 4/28/2025

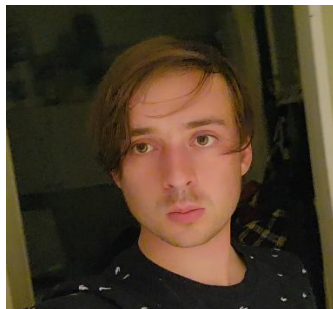
Members:



Cathy Wang



Kenneth Tseng



Micaiah Barletta



William Arida

Executive Summary

The project focused on the development of a 4 degree of freedom robotic arm designed to perform a series of competition tasks. Throughout the semester, the team collaborated to learn and apply skills in robotic design, kinematic analysis, and control programming. The final design, based on a closed four-bar linkage, was 3D printed using PLA plastic and reinforced with other materials hardware to meet the competition's weight and motion requirements. In the conceptual phase, each team member proposed independent designs, with the final choice based on a closed four bar linkage for enhanced stability. Using MATLAB, the team derived inverse kinematic equations, allowing them to control motions.

During the competition, the robotic arm successfully completed the first four rounds, which involved simple pick and drop tasks. However, the final and most complex task-picking up a pen, drawing a circle, and dropping a pen-was not successfully completed. Before the final task, the prototype suffered a mechanical failure when one of the motors broke, preventing the team from finishing the last challenge. As a result, the team finished in last place overall.

The team identified several potential contributing factors, including electrical problems and overheating, as well as software-related inaccuracies between commanded and actual positions. Despite these challenges, the project successfully demonstrated proof of concept. The arm was able to perform fundamental pick and drop operations under competition conditions, validating the design approach. The competition performance highlighted both the strengths and weaknesses of the initial design, particularly in mechanical durability under stress.

Introduction

The goal of this project was to design and build a robot with 4 degrees of freedom which could competently perform tasks in a competition. Throughout the semester the team learned a specific skillset which would allow them to collectively fabricate a working prototype. The process began with creating options for the eventual final design. The team took inspiration from a combination of design philosophy learned in class, as well as already existing robotic arm designs from research. For the control of the robotic arm, the team learned forward and inverse kinematic analysis, and practiced coding the mathematical process in Matlab. The hardware of the robotic arm was selected based on the competition task requirements of weight capacity and range of motion. The selected method of manufacture for the body of the robot was 3D printing PLA plastic, fastened with aluminum screws, bolts, and spacers.

The overall schematic design of the robotic arm had constraints on cost, but also needed proper weight distribution as well as ability withstand the forces placed on its components during movement. With this in mind, calculations were done to determine the maximum force exerted on the limbs and what thickness would allow for that to be properly distributed.

The mobility requirements of the robot arm were based on the competition tasks it was built to complete. This meant that it needed to have a workspace which was large enough to encompass the competition area, as well as allow for trajectories which would allow it to complete tasks without encountering obstacles or obstructions. For this reason, the team chose a combination of actuators; 1 stepper motor for rotation about the y axis (allowing for x axis movement), and 3 servo motors for the y/z plane movement and the gripper actuation. These 4 actuators allow for 4 degrees of freedom for the robotic arm, as per the motion requirements of the competition. As a whole, the team's selection of actuators implemented into their arm and

gripper design allowed for a functional prototype robotic arm which comfortably fit the criteria for the competition, allowing for effective completion of the tasks it was set to encounter.

Conceptual Design

For the conceptual design stage, each team member proposed an independent design. Cathy proposed a design built around a closed four-bar linkage. Figure 1 below shows the outline of the proposed linkage. The core design philosophy behind this design was that a closed four-bar linkage would likely impart the linkage with greater stability, as opposed to an open linkage. The design also featured an end gripper which would always be oriented horizontally. This horizontal orientation would be achieved through the kinematic design of the links rather than actuation. This approach has the advantage of reduced weight and less-intensive inverse kinematic calculations. The base of the four-bar linkage would rotate in the horizontal plane via a rotational actuator.

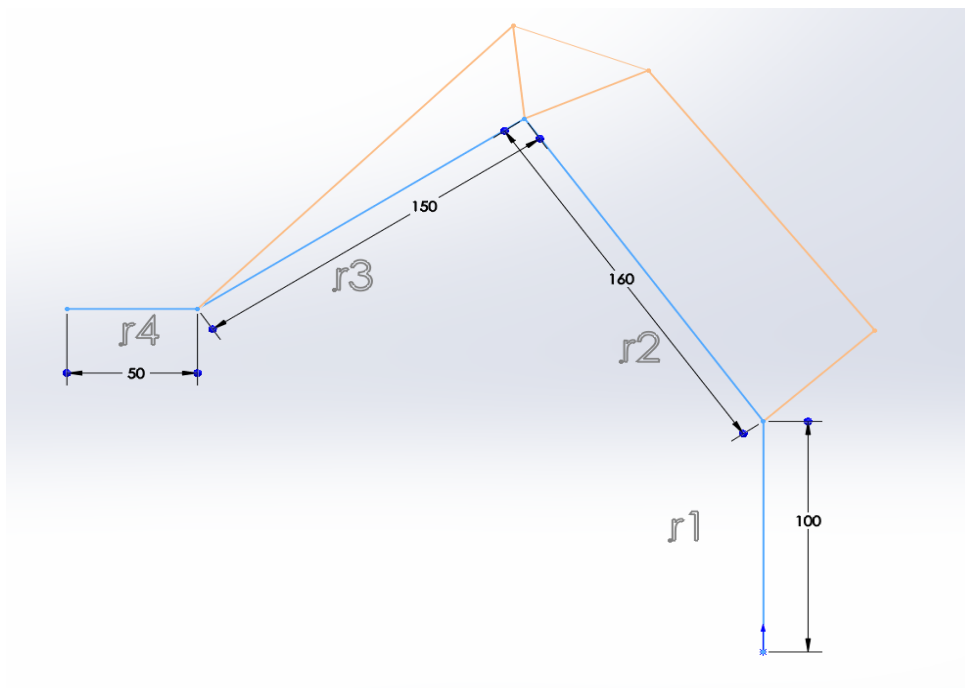


Figure 1: Cathy's Concept Design

Kenneth proposed an open-loop 4-bar linkage, shown below in figure 2. Here, the linkage would have a fixed base link and three free links, including the end-effector link. In contrast to the prior design, this design would feature an end-effector with a full degree of freedom rather than being fixed to one orientation. Additionally, the open-loop design would afford fewer constraints on the workspace, allowing for a broader range of linkage configurations. Again, the base link would be rotated in the horizontal plane via a rotational actuator.

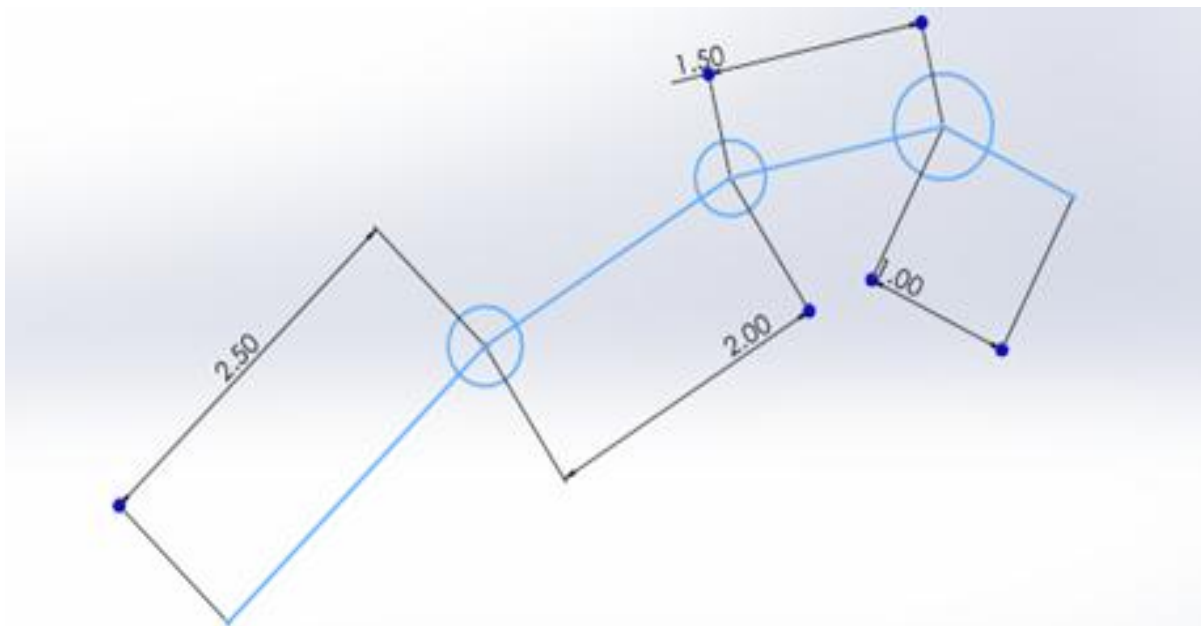


Figure 2: Kenneth's Concept Design

For Micaiah's design, shown in figure 3, a closed four-bar linkage was proposed. This design took inspiration from Cathy's design, but with the distinction of having an end effector with a full degree of freedom, rather than being kinematically constrained. Like the previous designs, the base would be actuated in the horizontal plane by situating a rotary actuator below the base link.

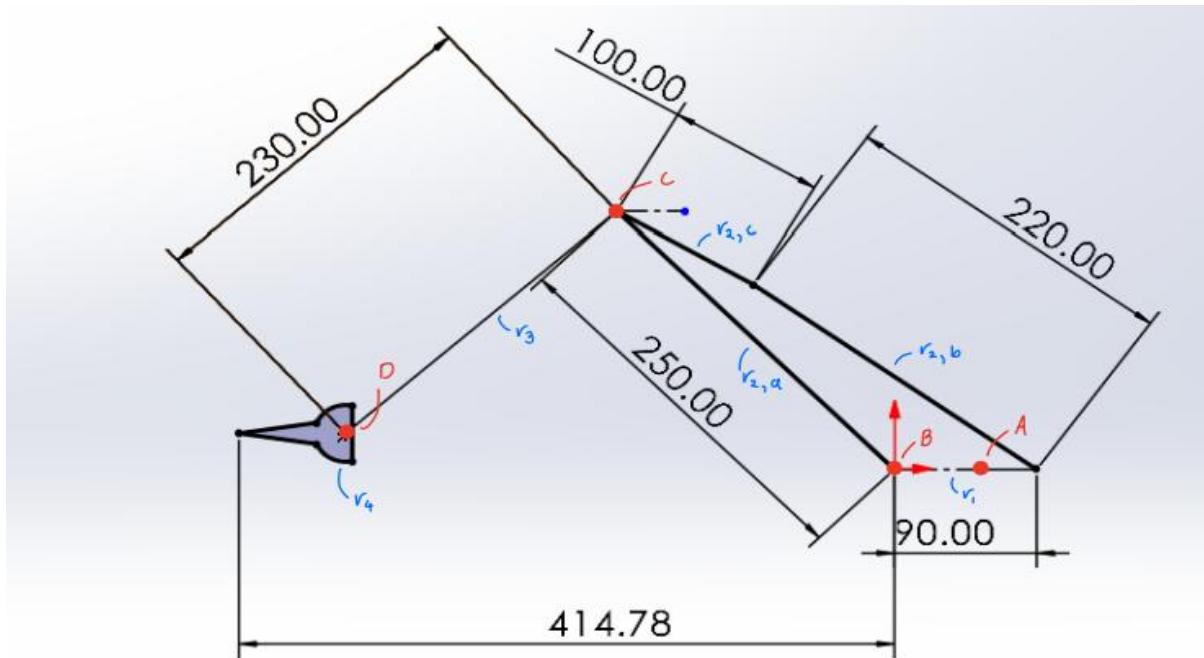


Figure 3: Micaiah's Concept Design

For Will's design, he proposed an open-loop design similar to Kenneth. Here again, the linkage would consist of one fixed base link, and three free links, giving a total of 4 degrees of freedom. The base link would be controlled by a rotary actuator, rotating it in the horizontal plane.

Having assessed each concept, the team decided that Cathy's proposed design was the best option to pursue. The team felt that this design had several key advantages over the others. First, unlike the open-loop, designs, the closed-loop design would likely provide more stability. Additionally, while the other designs proposed end effectors with full degrees of freedom, Cathy's design constrained the end effector to the horizontal. By doing so, the inverse kinematic calculations would be simplified and the end effector could be lighter, requiring less actuation.

That said, the team was also aware that pursuing this design would come with its drawbacks compared to the others. First, having a closed-loop design would restrict the workspace and the risk of link collision/conflict would be heightened. Additionally, having an end effector that is fixed to one direction would limit the ways in which the end-effector could

orient itself with respect to objects. Thus, the end effector orientation might be sub-optimal for certain tasks.

That noted, the team felt that the advantages of having greater stability and simpler inverse kinematics outweighed these concerns. So, Cathy's design was chosen to serve as the base for the final concept design.

Details of Final Design

Having selected Cathy's concept, the team refined this concept into their final design.

This design is shown below in figure 4.

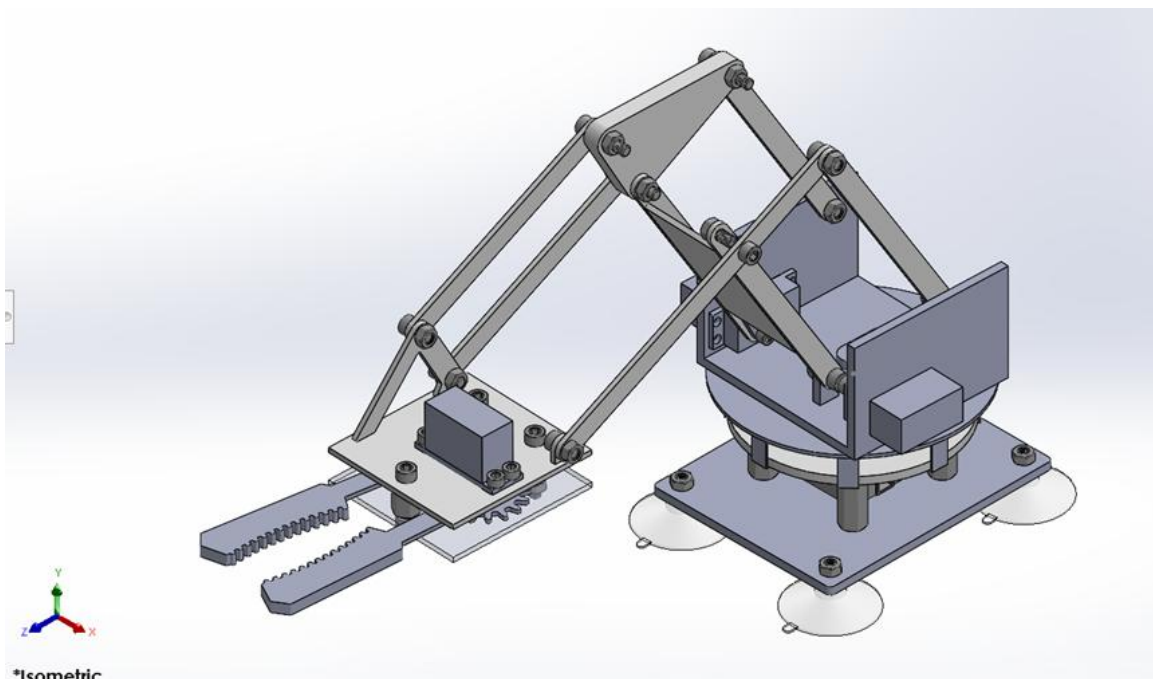


Figure 4: The Team's Final Design (CAD Model)

In keeping with Cathy's initial concept, the design features a closed-loop, 4-bar linkage. This closed linkage rests atop a base which is rotated by stepper motor. The four-bar linkage itself is actuated by two servo motors, which are able navigate the end effector to its desired position in the vertical plane. The team used suction cups on the very base of design to enhance

the arm's stability. Additionally, four bearings were situated below the rotated base. This helped to enhance balance and also reduce any nonaxial strain on the stepper motor shaft.

Taking a closer look at the end effector, featured a jaw-like design for grasping objects. As can be seen in figure 5, The gripper consisted of two adjacent jaws, which were integrated via gears, and actuated via a servo. The jaws featured “teeth” as well as some net curvature to enhance the gripper's ability to securely grasp objects.

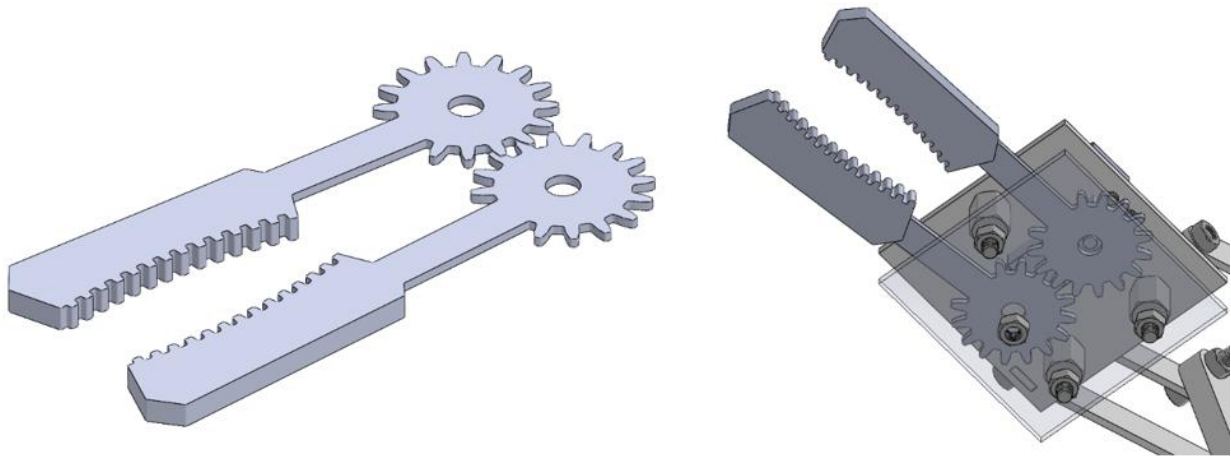


Figure 5: Gripper Design (CAD Model)

For the materials, the design consisted of a combination of printed parts and purchased parts. In figure 6 below, the bill of materials is provided. This bill of materials is not comprehensive but does include all components that were modelled in CAD.

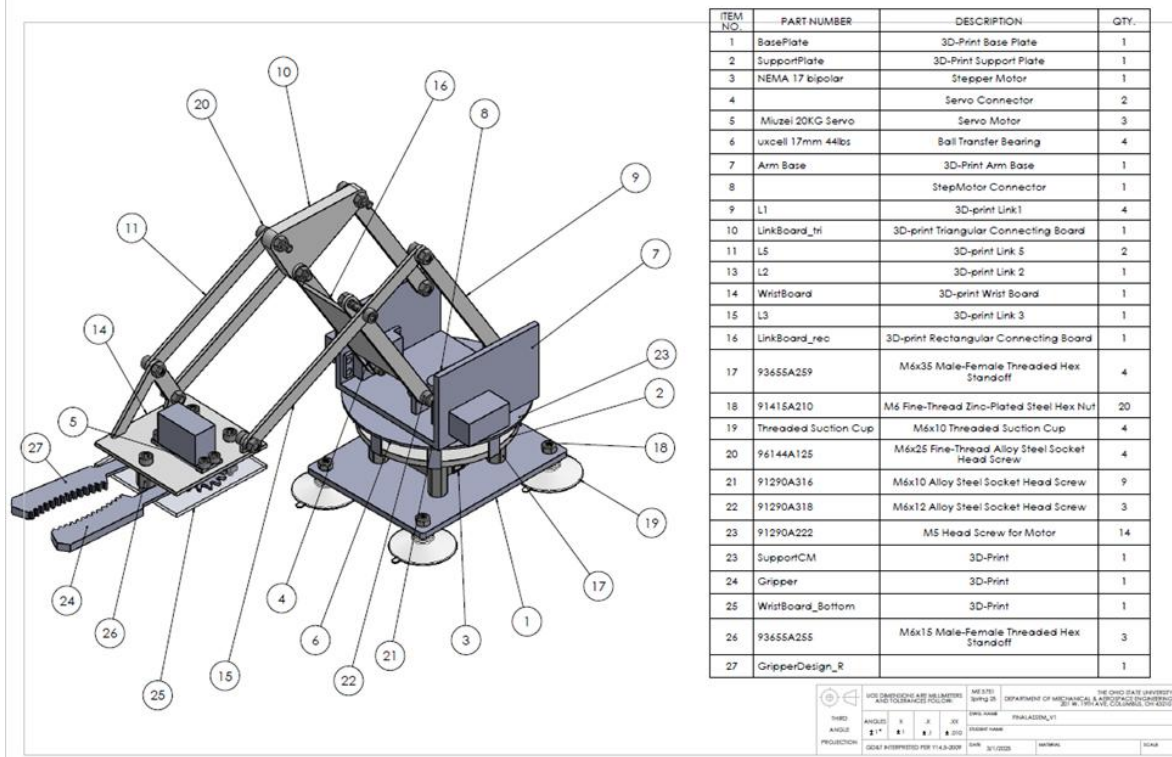


Figure 6: Bill of Materials for CAD Model

A more comprehensive list of the components used is provided in the following two tables. These tables are segregated by whether the parts were intended to be 3D printed or purchased.

Table 1: All 3D Printed Parts

Printable Parts		
Part Number	Description	#
1	Base Plate	1
2	Support Plate	1
7	Arm Base	1
9	Link 1	4

10	Triangular Link Board	1
11	Link 5	2
13	Link 2	1
14	Wrist Board	1
15	Link 3	1
16	Rectangular Link Board	1
23	Support CM	1
24	Gripper	2
25	Bottom Wrist Board	1

Table 2: Purchasable Parts for Design

Purchasable Parts				
<i>Part Number</i>	<i>Description</i>	<i>#</i>	<i>Cost</i>	<i>Component Type/function</i>
17	M6 x 35 Male-Female Threaded Hex Standoff	4	\$8.99	Attachment
26	M6 x 15 Male-Female Threaded Hex Standoff	3	\$6.33	Attachment
18	M6 Fine- Thread Zinc Plated Steel Hex Nut	20	\$15.99 (1 kit)	Attachment
20	M6 x 25 Fine Threaded Alloy Steel Socket Head Screw	4		Attachment
22	M6 x 12 Alloy Steel Socket Head Screw	3		Attachment
21	M6 x 10 Alloy Steel socket Head screw	9		Attachment
23	M5 Head Screw	14	\$8.99	Attachment
19	M6x 10 Threaded Suction Cup	4	\$12.50	Stabilization
3,8	NEMA 17 Bipolar Stepper Motor	1	\$13.99	Actuator (Stepper Motor)
5, 4	MiuZe 20KG Servo Motor	2	\$12.50*2	Actuator (Servo Motor)
†	DS3235 35 kg Servo	1	<i>Already Owned</i>	Actuator (Servo Motor)
6	Uxcell 17mm 44lbs Ball Transfer Bearing	4	\$12.79	Weight Distribution/Stabilization

†	Arduino Mega Board	1	<i>Already Owned</i>	Processor
†	Electrical Wiring Components (Resistors, Wires, etc.)	<i>N/A</i>	<i>Already Owned</i>	Wiring
†	Breadboard	1	<i>Already Owned</i>	Wiring
†	Buck Converter	1	\$18.34	Wiring
†	TB6600 Driver	1	\$10.89	Processing
†	Adjustable DC Power Supply	1	\$14.88	Wiring
†	Indicates a part not shown on the bill of materials			
Total Cost			\$148.69	

For the kinematic analysis, the team needed to determine equations for the angles actuated by the two servos situated at the base of the closed linkage. Specifically, these servos would be responsible for controlling links r_1 and r_3 , as labelled in the following figure. Thus, the team needed to determine mathematical equations for the corresponding angles θ_1 and θ_3 .

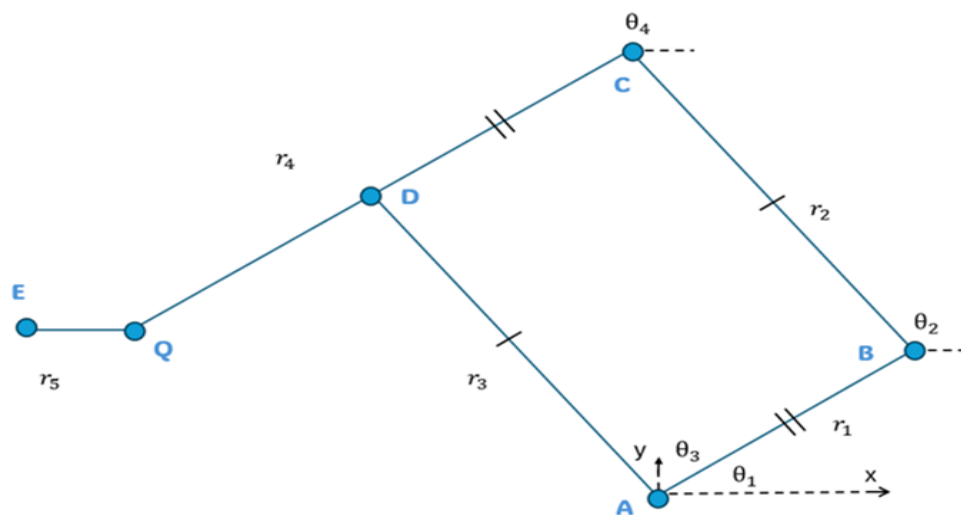


Figure 7: Kinematic Sketch of the Linkage

The team was able to successfully solve these angles by hand, producing the following set of equations:

$$\theta_1 = \arcsin\left(\frac{r_2^2 - Q_x^2 - QD^2 - Q_y^2}{\sqrt{(2 \times Q_x \times QD)^2 + (2 \times Q_y \times QD)^2}}\right) - \arctan\left(\frac{2 \times Q_x \times QD}{2 \times Q_y \times QD}\right)$$

$$\theta_3 = \theta_2 = \arccos\left(\frac{Q_x + QD \times \cos(\theta_1)}{r_2}\right)$$

Having solved the inverse kinematics, the team next needed to develop an algorithmic concept for using these equations to control the arm. Because the team was using a closed linkage, the team concluded that the team would have to approximate synchronous control of both base servos. Unlike in an open linkage, the issue of link collisions is a far greater concern for a closed linkage. Realistically, if for example, theta 3 were changed from 0° to 90° while theta 1 remained unchanged, well, then collisions would likely occur between the links. Thus, both servos would need to be controlled in concert if collisions were to be avoided.

Unfortunately, with the processor used, the Arduino MEGA, the team knew that true synchronous control of both servos was not feasible. This was due to how the Arduino MEGA is built. Consequently, the team decided to approximate synchronous control of both servos. To go about this, the team decided that a home position would be defined for the end effector. This would be the gripper's designated default position. To navigate from the home position to a desired task position, the team developed an algorithm to define a linear line between both points. Then, using the inverse kinematic equations, the angles corresponding to each discrete point within this line would be solved for. This would produce vectors of angles, one each for theta 1 and theta 3. These vectors could be used to gradually transition the gripper from its home position to a desired position.

To test whether this algorithm was feasible, it was implemented in MATLAB. Once the angles were solved for, they were then exported to SOLIDWORKS to complete a motion analysis.

The following image shows the linear line generated in MATLAB.

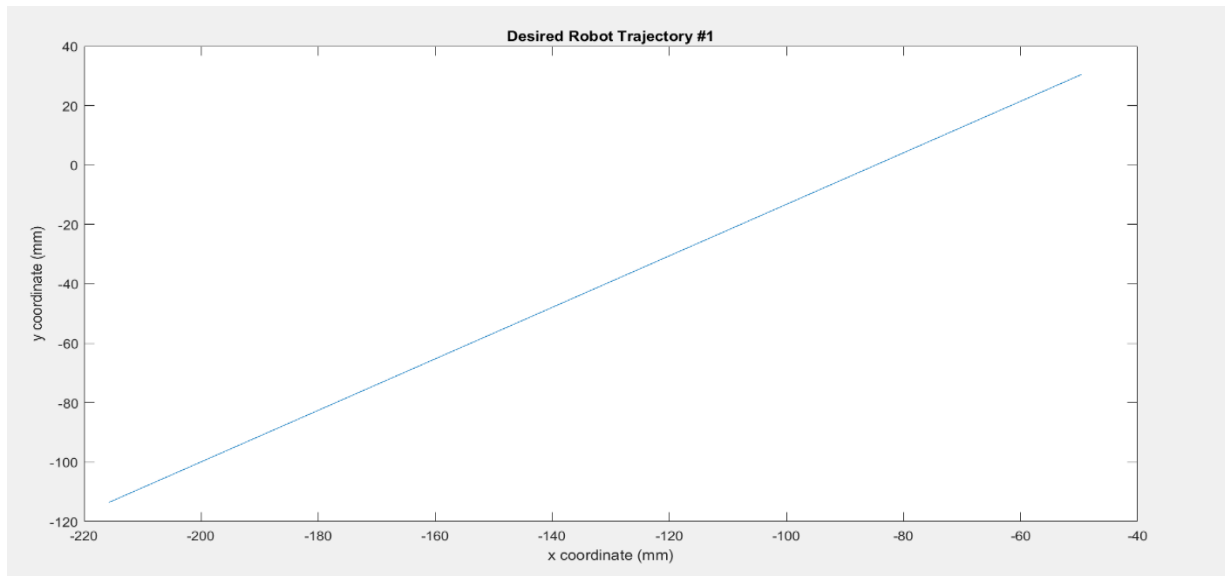


Figure 8: Linear Line for which the Corresponding Angles were Solved for

And, as seen in the next image, this is the trajectory the CAD model followed in a motion simulation, using the solved for angles.

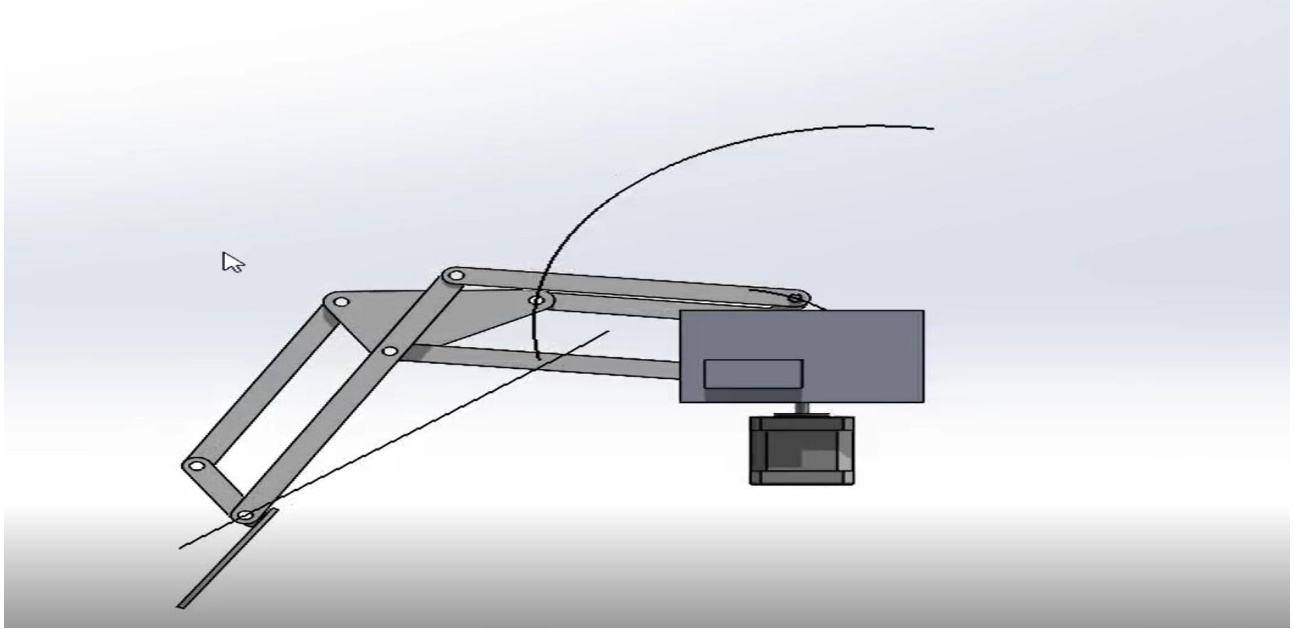


Figure 9: Trajectory of CAD Model in Motion Simulation

As shown in figure 9, a point on the end effector successfully followed the modelled linear line in the simulation. This confirmed both that the inverse kinematic equations were correct and that the proposed algorithm was functional.

The one limitation of using a linear line was that there might be cases where the end point is within the workspace, but some of the transitional points are not, rendering the desired motion impossible. With this concern in mind, the team modeled the workspace of the prototype using the Monte Carlo method. Shown below is the 3D workspace volume generated by the team in MATLAB. As will be discussed in the Testing and Competition Results section of this report though, there were likely some flaws in developing and utilizing this workspace.

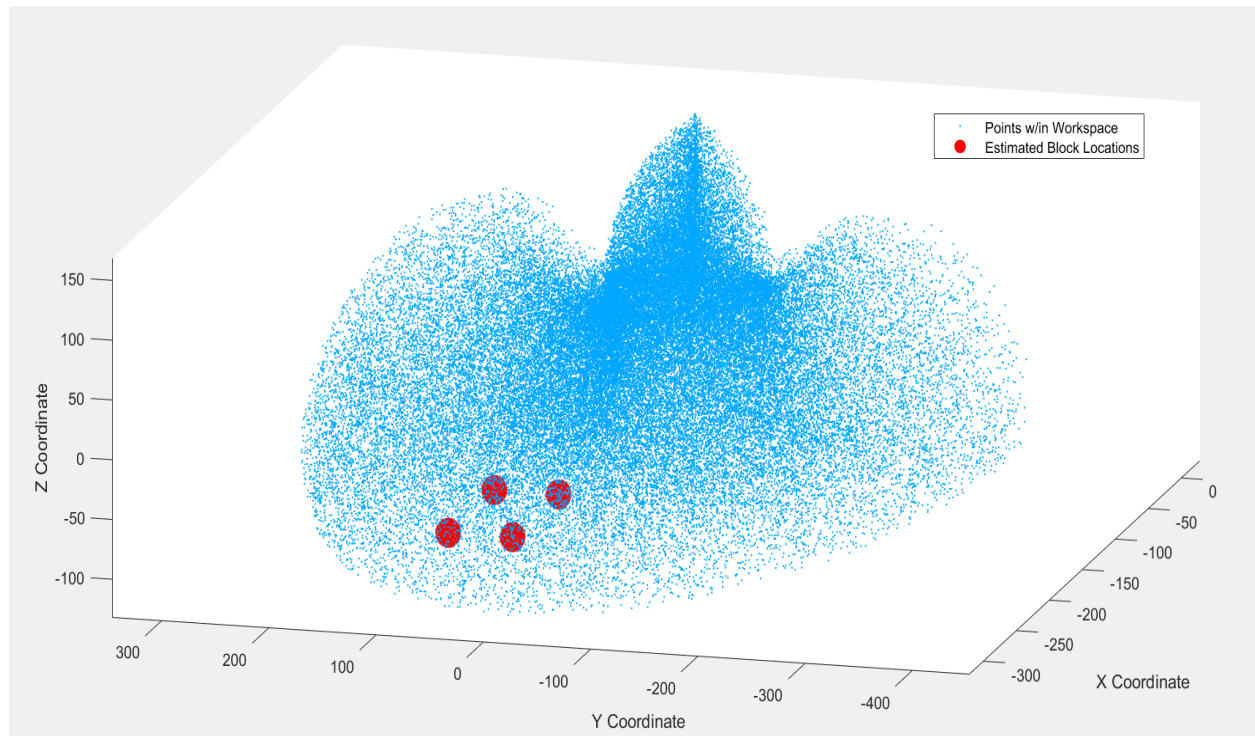


Figure 10: Workspace for Final Design

Fabrication Plan

The fabrication and assembly process for the prototype involved components purchasing, 3D printing, and assembling the robots based on the CAD model. To begin with, all required components, including motors, suction cups, and screws, were purchased from Amazon. For customized parts, the team utilized 3D printing. The white parts in the Figure () below were printed by the printers in the MAE department, while the remaining parts were printed by individuals who had access to a personal 3D printer.

The assembly process started with constructing the major arms. The white links were connected using M6 screws and washers, with instructions following the CAD model to ensure correct positioning. Additionally, the “ Δ ” shaped part, indicated by orange arrow in Figure 11, was inserted and glued onto the top wrist board. The purple rectangular board, indicated by the blue arrow in Figure 11, was also inserted and glued onto the two purple links on the sides.

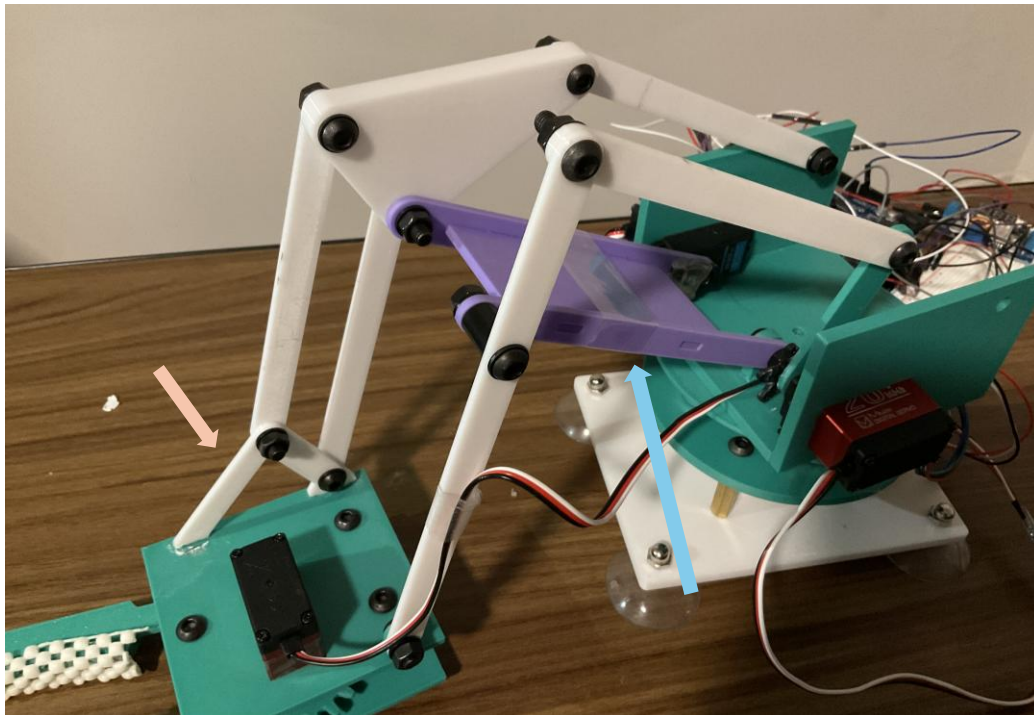


Figure 11: Prototype – Major Arm

Next, the servo motors were installed. Since the motors could not fit directly into the rectangular cutouts, they were first disassembled. After carefully inserting them into their designated locations, the motors were reassembled and secured. The servo motor locations are indicated below in Figure 12.

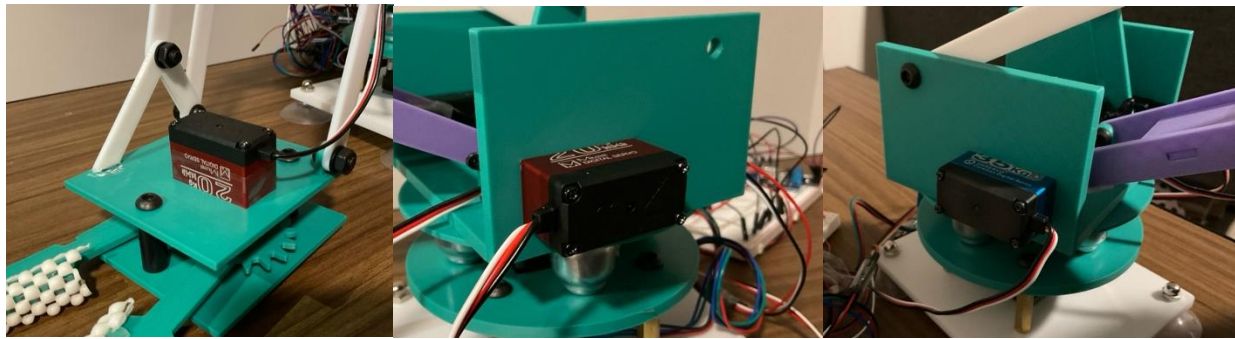


Figure 12: Prototype – Servo Motor Locations (Left: motor on the wrist board. Middle: motor on the right side of the base. Right: motor on the left side of the base.)

The next step was to assemble the base structure. 35 mm standoffs were attached to the rectangular (white) base plate, and the suction cups were also mounted at each of the four corners of the plate to provide stability.

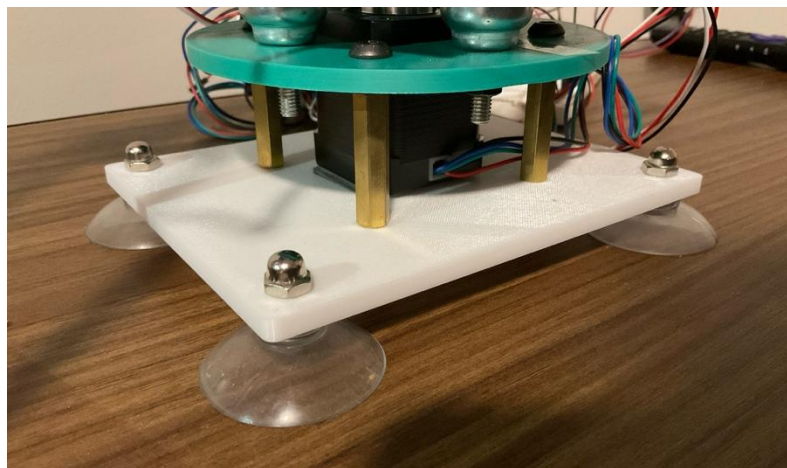


Figure 13: Prototype – Base Plate with Suction Cups and 35mm Standoffs

Ball bearings were then secured in their respective positions on the round plate, which is indicated in Figure 14. After that, the round plate with the bearings was attached to the standoffs

and fastened with M6 screws. The stepper motor was inserted into the square hole in the round plate and was fixed onto the white base plate.

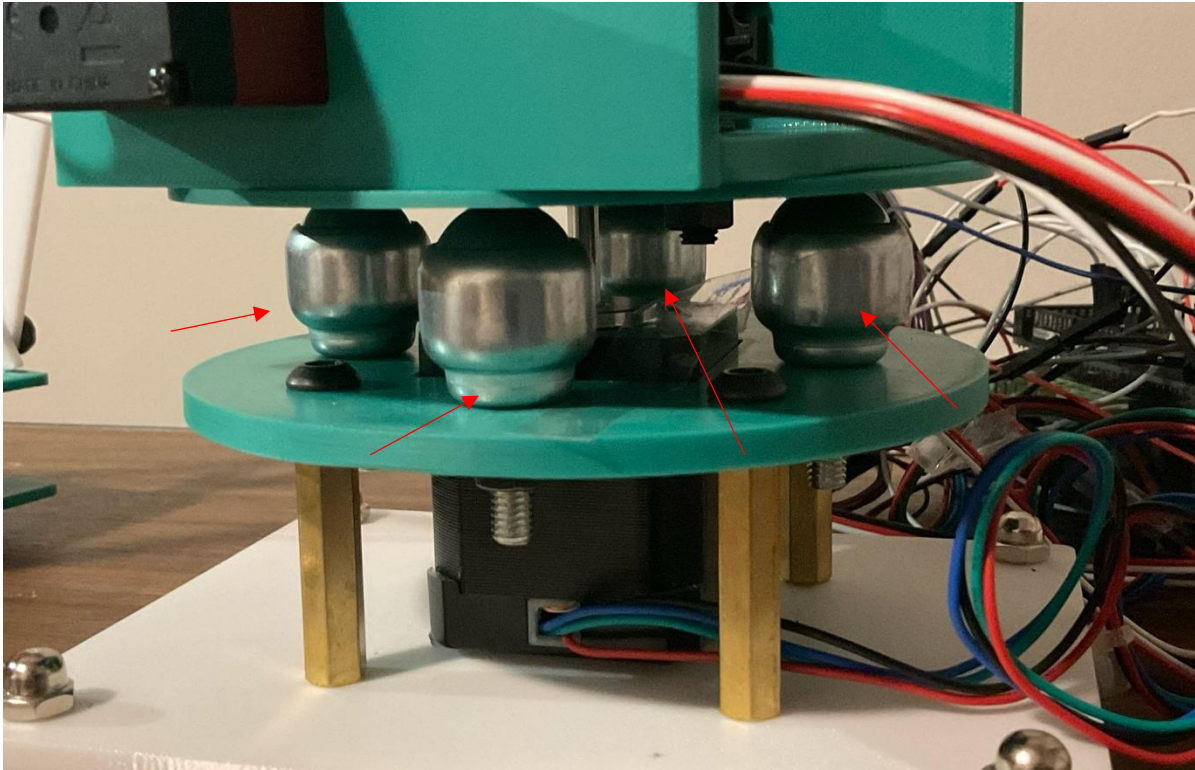


Figure 14: Prototype – Bearings (Indicated by Red Arrows) and Step Motor

Following this, the U-shaped base was connected to a thin round plate and secured with screws. A small 5mm diameter hole in both of them was used to insert the stepper motor driver. The hole had a tight clearance so that the connection is stable enough to avoid the need of additional fastening. The connection is shown in Figure 15.

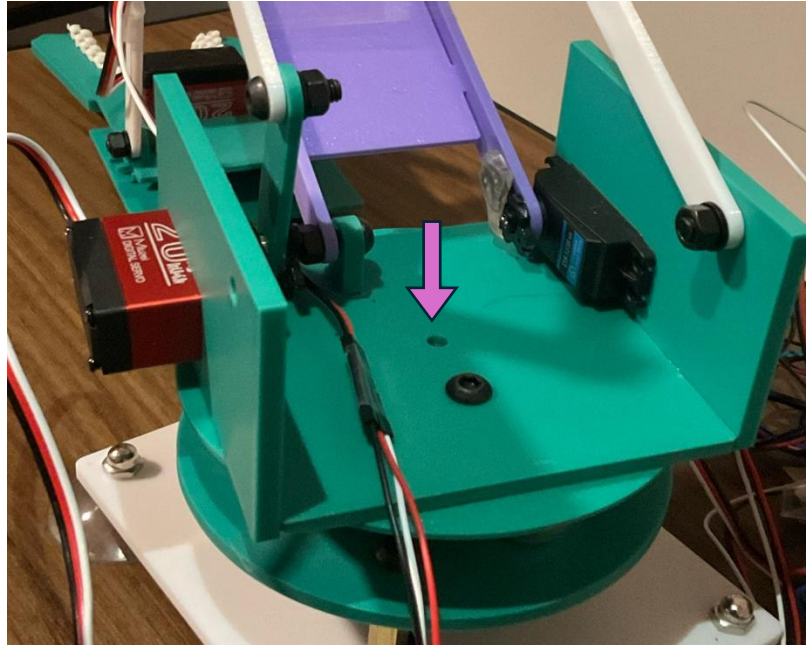


Figure 15: Prototype – U-shape Base and Thin Round Plate, with the indicated 5mm hole for the step motor.

The gripper assembly involved the following steps. The right gripper, shown on the right in Figure 16, was attached directly to a servo motor that's already attached to the top wrist plate. The left gripper, shown on the left in Figure 16, was mounted to the bottom wrist plate with a 5mm standoff in between to ensure proper alignment between the two grippers. Then, the top and bottom wrist plates are connected with M6 screws and 20 mm standoff.



Figure 16: Prototype – Gripper Assembly

Finally, the major arm links were connected to the base assembly using M6 screws. This completes the structural assembly of the prototype, following the design outlined in the CAD model. The completed prototype is shown in Figure 17.

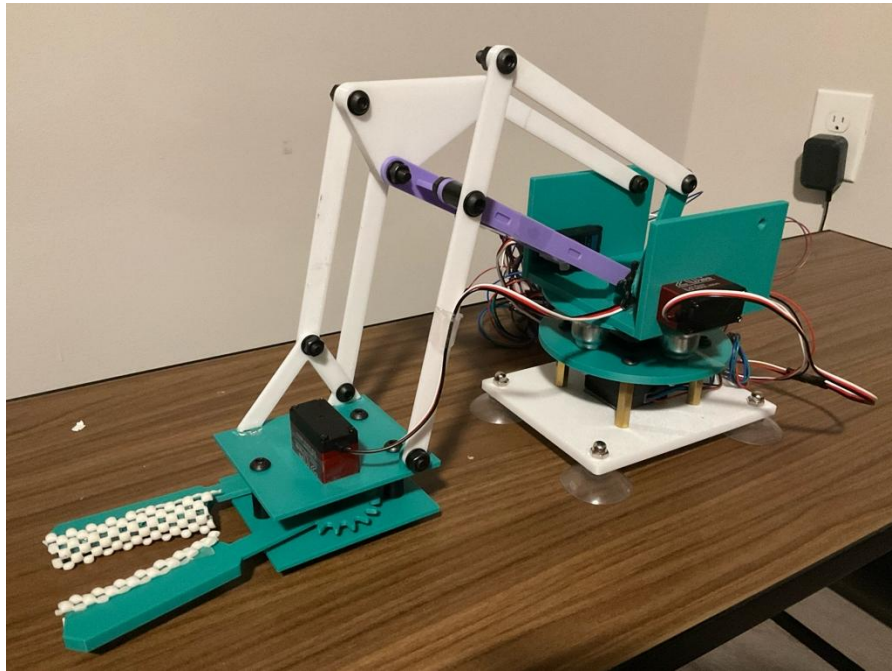


Figure 17: Prototype – Completed Prototype

Testing and Competition Results

During the testing of the prototype, the prototype proved to have sufficient speed and payload capacity. While no measurement of the prototype's actual payload capacity was obtained, it was calculated to be approximately 1.2 kilograms (see appendix for calculation), which was more than sufficient for the competition. During testing, the prototype did not encounter any difficulties with the weight of either the blocks or the pen used during testing, thus lending some validity to the calculated payload capacity.

In terms of speed, the prototype was able to complete tasks with the allotted time limit, although it skewed toward the upper limits. For the block-pickup tasks, the tasks were able to be completed with approximately 60 to 70 seconds, thus meeting the 90 second limit. The same

goes for the circle-tracing task, which averaged around 100 to 110 seconds, again meeting the time limit – 120 seconds.

Unfortunately, during the competition, the robot's performance did not meet the team's expectations. For the block-pick up tasks, while some tests runs were successful, the robot lacked consistency. During the competition, the robot struggled to consistently pick up blocks, sometimes missing the mark. Additionally, during testing, multiple issues were encountered while trying to pick up the pen and trace a circle. During the competition, the robot incurred damage to some of its key systems, rendering it inoperable for the last task.

In reflecting on the prototype's underwhelming performance, several critical factors that likely contributed have been identified.

From a hardware perspective, there was one clear weak point the design suffered from. In order to attach the servos to the actuated links, the team used attachable servo gears that came with the servos when they were purchased. Unfortunately, the plastic these servo gears were made of was likely too soft and pliable. During operation, the mesh between the servo shaft and the attachable servo gears would start to wear. Multiple times during testing, the mesh had become completely stripped, meaning the servo could no longer actuate the link. This had led to the prototype needing to be repeatedly disassembled to attach new servo gears. Unfortunately, this design flaw was realized only a few days prior to the competition, making it difficult to find an alternative.

Other issues arose from the electrical wiring of the prototype. First, one oddity was that the servos would experience "jolts" where they would randomly jump to an unwritten position. (Interestingly, this "jolting" behavior was not observed in the stepper.) This would occur occasionally during testing, particularly shortly after the code had been uploaded to the

prototype. The exact cause for this is unknown, but the team speculates that this behavior was the result of voltage spikes. While a buck converter was used to ensure that the voltage was reduced from the DC supply to an appropriate range for the servos, it is also possible the buck converter's displayed output voltage was not entirely accurate. Perhaps it was supplying too much voltage to the servos. It is also possible there was a charge build up occurring.

Another electrical issue was also observed. During the competition, one of the servos became extremely hot. This excess heat was not observed in prior testing. It is possible the servo had in some way been damaged, either physically or electrically. For example, a voltage spike may have damaged the servo's internal circuitry. Alternatively, some of the wiring may have been disrupted, potentially causing a short circuit, for example. When this "jolting" would occur, it was very rough on the prototype, and it likely damaged it to some extent.

Lastly, there were several critical issues with the software programming. First, there was the issue of the commanded position not corresponding to the observed position. For example, if the end effector were commanded to go 300 millimeters forward in the code, it might only go say, 200 millimeters forward. This meant that the team had to spend a considerable amount of time finding appropriate coordinates for the task positions, rather than being able to measure them. The exact cause of this issue is still unknown. Interestingly, this issue was not encountered in the MATLAB code. But when this same code was implemented into Arduino, that is when the issue manifested. It is possible there was a coding error made during this transition between programming languages. The servo motors were also imprecise and inaccurate, which likely contributed to this, although this likely was not the sole determining cause.

Another issue was that, even if a kinematic angle was out of the workspace, it would still be written to the servos. In cases when this would occur, it would inevitably lead to link

collisions and unnecessary strain and wear on the prototype. This would have been more avoidable if the written position corresponded to the actual position; however, this aforementioned discrepancy led to the team being unsure whether a given written position would be within the workspace.

Lastly, during testing, the software that controlled pen retrieval was not working as intended. The primary issue here was that, once the pen had been grasped in the holder, it could not rise vertically high enough to clear the pen container. Why exactly this issue was occurring is still unknown, but it is likely related to the prior issue of there being a discrepancy between the written and actual positions. This certainly is not comprehensive, but these are some of the primary issues that the prototype encountered.

So, given the opportunity to improve the design, what changes would the team implement to remedy these discussed issues? Starting from the hardware, the team could find more durable servo gear attachments. The weak plastic simply is not suitable. Going beyond this, it may be beneficial to design custom gear pairs for each servo. Not only would this allow for a stronger material to be used, but, if two gears are used per servo, the team could use gear ratios to amplify the torque coming from the servos. This would likely help to make the arm's motion steadier and more fluid. Additionally, having a properly meshed pair of gears would likely help lessen some of the "jittering" that was observed.

Moving on to the electrical issues, first, the team should verify with a multimeter whether the displayed voltage on the buck converter is accurate. Additionally, changing the electrical circuitry may be beneficial. For example, perhaps including a capacitor in the servo circuitry could prevent the seeming voltage spikes. Also, the wiring is very disorderly, making it hard to verify if all the wires are still in place. So, the organization of the wiring should be improved.

Lastly, the code could be enhanced. The issue causing the discrepancy between the written and actual positions should be identified. Additionally, if possible, a way to ensure that only angle pairs within the workspace are written to the servos should be developed. This could help to avoid unnecessary stress on the system.

Overall, while the design has many shortcomings, the team believes their design at very least demonstrates that their proof of concept is viable. Although it has its limitations, the core systems of the prototype proved to be functional during testing. So, while there is significant room for refinement, the team feels that they put forward a strong effort.

Appendix

Max Payload Calculation

m_p is the mass of the max payload

m_L is the estimated mass of the links

L is the approximate length from base at most extended position

g is gravity

T is torque of both base servos, as provided by data sheet

$$m_p = \frac{T}{g \times L} - m_L = \frac{5.88 \text{ Nm}}{\left(9.81 \frac{\text{m}}{\text{s}^2}\right) \times (0.35 \text{ m})} - 0.5 \text{ kg} = 1.212 \dots \text{ kg}$$